

Saatvik Palli

pallisaatvik@gmail.com | 614-705-3222 | [Portfolio](#) | [LinkedIn](#) | [GitHub](#)

EDUCATION

University of Illinois Urbana-Champaign

Master of Computer Science

Champaign, IL

Aug. 2026 – Dec. 2027

Michigan State University

B.S. in Computer Science, GPA: 3.9/4.0

East Lansing, MI

Aug. 2022 – May 2026

EXPERIENCE

Chief Technology Officer

Nexrup Music Discovery ↗

Aug. 2025 – Present

East Lansing, MI

- Enabled week-over-week momentum ranking for **144 independent artists** by building a daily ETL pipeline accumulating 380,000+ datapoints across 10 streaming and social platforms.
- Cut artist leaderboard load times from **3-4 seconds to ~600 ms** by shrinking the initial bundle 57%, replacing $O(n^2)$ scans with $O(1)$ hash-set lookups, and deduplicating concurrent API requests behind a TTL cache.
- Defined the data model and service boundaries while mentoring **3 developers and 2 designers** over 9 months through branch-per-developer workflow, code review, and 110+ PRs.

Data Analytics & Engineering Intern

Auto-Owners Insurance

May 2024 – Aug. 2026

Lansing, MI

- Spearheaded upgrading licensing management to an internal SQL Server system, saving **\$115K over 5 years**, by engineering ETL pipelines and stored procedures that surfaced real-time compliance status.
- Developed a neural network capable of scoring **10,000+ claims per second**, producing real-time difficulty ratings used to balance caseloads across adjusters, with scikit-learn and MLflow.
- Built a Power BI scoring model ranking **500+ adjusters across 43 branches** on **8 weighted claim-handling metrics**, exposing systematic gaps in rep training and technology adoption.
- Analyzed thousands of claims with Pandas and NumPy to project 18-month cost savings of chatbot-assisted salvage handling, generating reports that strengthened vendor negotiations and contract renewal.

Software Developer

Magna International, Michigan State Capstone

Aug. 2025 – Dec. 2025

East Lansing, MI

- Developed a CAD classification pipeline enabling engineers to find functionally similar parts across **2,000+ automotive components**, by tagging 3D meshes with a vision LLM using extracted geometry statistics.
- Increased tagging accuracy by 64%** through prompt A/B testing and multi-view rendering optimization.
- Cut per-request API cost by ~50%** and eliminated rate-limit failures by moving tagging to an OpenAI Batch API pipeline that chunked 1,800 parts into 90 asynchronous jobs with state persistence for interrupted runs.

Executive Board

Imagine Software Club at MSU

Aug. 2024 – May 2026

East Lansing, MI

- Interviewed and matched **~50 students to 6 startup projects**, aligning technical experience to effective teams.
- Designed and hosted a Bash scripting workshop for **60+ students** by hosting a custom server environment.

PROJECTS

Half Full Calendar ↗ | *JavaScript, Electron, Firebase, Google Calendar API*

- Built two-way Google Calendar synchronization backed by Firebase persistence, enabling offline access and cross-session state for a desktop calendar since released on the **Mac App Store and Microsoft Store**.
- Contributed frontend fixes to event creation and viewing flows on a team-built Electron application.

Insurance Risk Factor Neural Network | *Python, Flask, Scikit-Learn, Plotly, Git/GitHub*

- Led a team of 3 to build a web app visualizing health risk factors and projected financial loss.
- Improved model accuracy by 8%** on insurance dataset by tuning a scikit-learn MLP.

TECHNICAL SKILLS

Languages: Python, SQL, Bash, JavaScript, C++, C, DAX, HTML/CSS

Technologies: SQL Server, GCP, Docker, React, Flask, Firebase/Firestore, Pandas, NumPy, PyTorch, Scikit-learn, MLflow, Power BI, Plotly, Git/GitHub, Electron

Concepts: Data Modeling, ETL, Data Warehousing, Distributed Systems, Database Systems (SQL & NoSQL), Query Optimization, Algorithms & Data Structures, Machine Learning, Linux

Interests: Jazz trumpet, Eagle Scout, Brazilian jiu-jitsu, poker